## Chapter 4

**2.** $f(x) = \frac{1}{10}$ for $-5 \le x \le 5$ and $= 0$ otherwise

**a.** $P(X < 0) = \int_{-5}^{0} \frac{1}{10} dx = .5$.

**b.** $P(-2.5 < X < 2.5) = \int_{-2.5}^{2.5} \frac{1}{10} dx = .5$.

**c.** $P(-2 \le X \le 3) = \int_{-2}^{3} \frac{1}{10} dx = .5$.

**d.** $P(k < X < k+4) = \int_{k}^{k+4} \frac{1}{10} dx = \frac{1}{10} x \Big]_{k}^{k+4} = \frac{1}{10}[(k+4) - k] = .4$.

**4.**

**a.** $\int_{-\infty}^{\infty} f(x;\theta)dx = \int_{0}^{\infty} \frac{x}{\theta^2} e^{-x^2/2\theta^2} dx = -e^{-x^2/2\theta^2}\Big]_{0}^{\infty} = 0 - (-1) = 1$

**b.** $P(X \le 200) = \int_{-\infty}^{200} f(x;\theta)dx = \int_{0}^{200} \frac{x}{\theta^2} e^{-x^2/2\theta^2} dx = -e^{-x^2/2\theta^2}\Big]_{0}^{200} \approx -.1353 + 1 = .8647$.

$P(X < 200) = P(X \le 200) \approx .8647$, since $X$ is continuous.

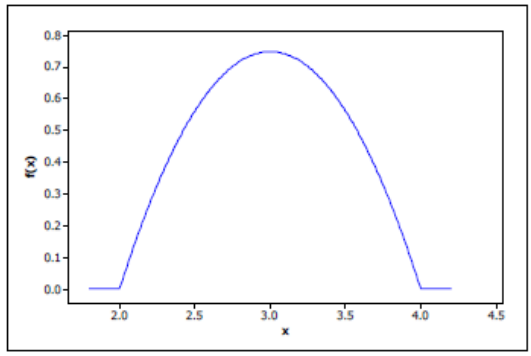
$P(X \ge 200) = 1 - P(X < 200) \approx .1353$.

**c.** $P(100 \le X \le 200) = \int_{100}^{200} f(x;\theta)dx = -e^{-x^2/20{,}000}\Big]_{100}^{200} \approx .4712$.

**d.** For $x > 0$, $P(X \le x) = \int_{-\infty}^{x} f(y;\theta)dy = \int_{0}^{x} \frac{y}{\theta^2} e^{-y^2/2\theta^2} dx = -e^{-y^2/2\theta^2}\Big]_{0}^{x} = 1 - e^{-x^2/2\theta^2}$.

**6.**

**a.**




**b.** $1 = \int_2^4 k[1-(x-3)^2]dx = \int_{-1}^1 k[1-u^2]du = \cdots = \dfrac{4k}{3} \Rightarrow k = \dfrac{3}{4}$.

**c.** $P(X > 3) = \int_3^4 \frac{3}{4}[1-(x-3)^2]dx = .5$. This matches the symmetry of the pdf about $x = 3$.

**d.** $P\left(\frac{11}{4} \le X \le \frac{13}{4}\right) = \int_{11/4}^{13/4} \frac{3}{4}[1-(x-3)^2]dx = \frac{3}{4}\int_{-1/4}^{1/4}[1-u^2]du = \dfrac{47}{128} \approx .367$.

**e.** $P(|X-3| > .5) = 1 - P(|X-3| \le .5) = 1 - P(2.5 \le X \le 3.5) =$
$1 - \int_{-.5}^{.5} \frac{3}{4}[1-u^2]du = \cdots = 1 - .6875 = .3125$.

**12.**

**a.** $P(X < 0) = F(0) = .5$.

**b.** $P(-1 \le X \le 1) = F(1) - F(-1) = .6875$.

**c.** $P(X > .5) = 1 - P(X \le .5) = 1 - F(.5) = 1 - .6836 = .3164$.

**d.** $f(x) = F'(x) = \dfrac{d}{dx}\left(\dfrac{1}{2} + \dfrac{3}{32}\left(4x - \dfrac{x^3}{3}\right)\right) = 0 + \dfrac{3}{32}\left(4 - \dfrac{3x^2}{3}\right) = .09375\left(4 - x^2\right)$.

**e.** By definition, $F(\tilde{\mu}) = .5$. $F(0) = .5$ from **a** above, which is as desired.

**14.**

**a.** If $X$ is uniformly distributed on the interval from $A$ to $B$, then $E(X) = \int_A^B x \cdot \frac{1}{B-A} dx = \frac{A+B}{2}$, the midpoint of the interval. Also, $E(X^2) = \frac{A^2 + AB + B^2}{3}$, from which $V(X) = E(X^2) - [E(X)]^2 = \ldots = \frac{(B-A)^2}{12}$.

With $A = 7.5$ and $B = 20$, $E(X) = 13.75$ and $V(X) = 13.02$.

**b.** From Example 4.6, the complete cdf is $F(x) = \begin{cases} 0 & x < 7.5 \\ \frac{x-7.5}{12.5} & 7.5 \le x < 20 \\ 1 & 20 \le x \end{cases}$.

**c.** $P(X \le 10) = F(10) = .200$; $P(10 \le X \le 15) = F(15) - F(10) = .4$.

**d.** $\sigma = \sqrt{13.02} = 3.61$, so $\mu \pm \sigma = (10.14, 17.36)$. Thus, $P(\mu - \sigma \le X \le \mu + \sigma) = P(10.14 \le X \le 17.36) = F(17.36) - F(10.14) = .5776$.
Similarly, $P(\mu - 2\sigma \le X \le \mu + 2\sigma) = P(6.53 \le X \le 20.97) = 1$.

**18.** $f(x) = \frac{1}{1-(-1)} = \frac{1}{2}$ for $-1 \le x \le 1$

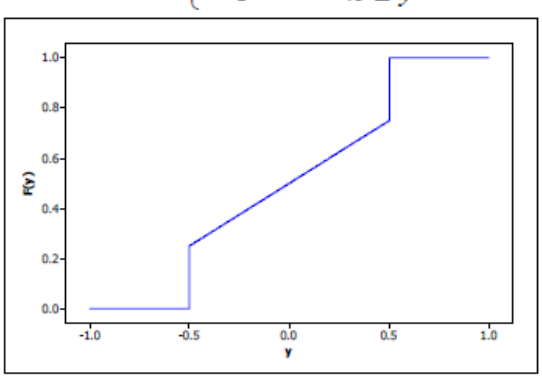
**a.** $P(Y = .5) = P(X \ge .5) = \int_{.5}^{1} \frac{1}{2} dx = .25$.

**b.** $P(Y = -.5) = .25$ as well, due to symmetry.

For $-.5 < y < .5$, $F(y) = .25 + \int_{-.5}^{y} \frac{1}{2} dx = .25 + .5(y + .5) = .5 + .5y$.

Since $Y \le .5$, $F(y) = 1$ for all $y \ge .5$. That is,

$$F(y) = \begin{cases} 0 & y < -.5 \\ .5 + .5y & -.5 \le y < .5 \\ 1 & .5 \le y \end{cases}$$



**22.**

**a.** For $1 \le x \le 2$, $F(x) = \int_1^x 2\left(1-\frac{1}{y^2}\right)dy = 2\left(y+\frac{1}{y}\right)\Big]_1^x = 2\left(x+\frac{1}{x}\right)-4$, so the cdf is

$$F(x) = \begin{cases} 0 & x<1 \\ 2\left(x+\frac{1}{x}\right)-4 & 1 \le x \le 2 \\ 1 & x>2 \end{cases}$$

**b.** Set $F(x) = p$ and solve for $x$: $2\left(x+\frac{1}{x}\right)-4 = p \Rightarrow 2x^2-(p+4)x+2=0 \Rightarrow$

$\eta(p) = x = \frac{(p+4)+\sqrt{(p+4)^2-4(2)(2)}}{2(2)} = \frac{p+4+\sqrt{p^2+8p}}{4}$. (The other root of the quadratic gives solutions outside the interval [1, 2].) To find the median $\tilde{\mu}$, set $p = .5$: $\tilde{\mu} = \eta(.5) = \ldots = 1.640$.

**c.** $E(X) = \int_1^2 x \cdot 2\left(1-\frac{1}{x^2}\right)dx = 2\int_1^2\left(x-\frac{1}{x}\right)dx = 2\left(\frac{x^2}{2}-\ln(x)\right)\Big]_1^2 = 1.614$. Similarly,

$E(X^2) = 2\int_1^2\left(x^2-1\right)dx = 2\left(\frac{x^3}{3}-x\right)\Big]_1^2 = \frac{8}{3} \Rightarrow V(X) = .0626.$

**d.** The amount left is given by $h(x) = \max(1.5-x, 0)$, so

$E(h(X)) = \int_1^2 \max(1.5-x,0)f(x)dx = 2\int_1^{1.5}(1.5-x)\left(1-\frac{1}{x^2}\right)dx = .061$.

**30.**

**a.** $\Phi(c) = .9100 \Rightarrow c \approx 1.34$, since .9099 is the entry in the 1.3 row, .04 column.

**b.** Since the standard normal distribution is symmetric about $z = 0$, the 9th percentile = –[the 91st percentile] = –1.34.

**c.** $\Phi(c) = .7500 \Rightarrow c \approx .675$, since .7486 and .7517 are in the .67 and .68 entries, respectively.

**d.** Since the standard normal distribution is symmetric about $z = 0$, the 25th percentile = –[the 75th percentile] = –.675.

**e.** $\Phi(c) = .06 \Rightarrow c \approx -1.555$, since .0594 and .0606 appear as the –1.56 and –1.55 entries, respectively.

**32.**

**a.** $P(X \le 15) = P\left(Z \le \frac{15-15.0}{1.25}\right) = P(Z \le 0) = \Phi(0.00) = .5000.$

**b.** $P(X \le 17.5) = P\left(Z \le \frac{17.5-15.0}{1.25}\right) = P(Z \le 2) = \Phi(2.00) = .9772.$

**c.** $P(X \ge 10) = P\left(Z \ge \frac{10-15.0}{1.25}\right) = P(Z \ge -4) = 1 - \Phi(-4.00) = 1 - .0000 = 1.$

**d.** $P(14 \le X \le 18) = P\left(\frac{14-15.0}{1.25} \le Z \le \frac{18-15.0}{1.25}\right) = P(-.8 \le Z \le 2.4) = \Phi(2.40) - \Phi(-0.80) = .9918 - .2119 = .7799.$

**e.** $P(|X - 15| \le 3) = P(-3 \le X - 15 \le 3) = P(12 \le X \le 18) = P(-2.4 \le Z \le 2.4) = \Phi(2.40) - \Phi(-2.40) = .9918 - .0082 = .9836.$

**34.** $\mu = .30$, $\sigma = .06$

**a.** $P(X > .50) = P(Z > 3.33) = 1 - \Phi(3.33) = 1 - .9996 = .0004.$

**b.** $P(X \le .20) = \Phi(-0.50) = .3085.$

**c.** We want the 95th percentile, $c$, of this normal distribution, so that 5% of the values are higher. The 95th percentile of the standard normal distribution satisfies $\Phi(z) = .95$, which from the normal table yields $z = 1.645$.

So, $c = .30 + (1.645)(.06) = .3987$. The largest 5% of all concentration values are above .3987 mg/cm$^3$.

**44.**

**a.** $P(\mu - 1.5\sigma \le X \le \mu + 1.5\sigma) = P(-1.5 \le Z \le 1.5) = \Phi(1.50) - \Phi(-1.50) = .8664.$

**b.** $P(X < \mu - 2.5\sigma \text{ or } X > \mu + 2.5\sigma) = 1 - P(\mu - 2.5\sigma \le X \le \mu + 2.5\sigma)$
$= 1 - P(-2.5 \le Z \le 2.5) = 1 - .9876 = .0124.$

**c.** $P(\mu - 2\sigma \le X \le \mu - \sigma \text{ or } \mu + \sigma \le X \le \mu + 2\sigma) = P(\text{within 2 sd's}) - P(\text{within 1 sd}) =$
$P(\mu - 2\sigma \le X \le \mu + 2\sigma) - P(\mu - \sigma \le X \le \mu + \sigma) = .9544 - .6826 = .2718.$

**50.** We use a normal approximation to the binomial distribution: Let $X$ denote the number of people in the sample of 1000 who <u>can</u> taste the difference, so $X \sim \text{Bin}(1000, .03)$. Because $\mu = np = 1000(.03) = 30$ and $\sigma = \sqrt{np(1-p)} = 5.394$, $X$ is approximately $N(30, 5.394)$.

**a.** Using a continuity correction, $P(X \ge 40) = 1 - P(X \le 39) = 1 - P\left(Z \le \frac{39.5-30}{5.394}\right) =$
$1 - P(Z \le 1.76) = 1 - \Phi(1.76) = 1 - .9608 = .0392.$

**b.** 5% of 1000 is 50, and $P(X \le 50) = P\left(Z \le \frac{50.5-30}{5.394}\right) = \Phi(3.80) \approx 1.$

**54.** Use the normal approximation to the binomial, with a continuity correction. With $p = .10$ and $n = 200$, $\mu = np = 20$, and $\sigma^2 = npq = 18$. So, Bin(200, .10) $\approx N(20, \sqrt{18})$.

**a.** $P(X \le 30) = \Phi\left(\dfrac{(30+.5)-20}{\sqrt{18}}\right) = \Phi(2.47) = .9932.$

**b.** $P(X < 30) = P(X \le 29) = \Phi\left(\dfrac{(29+.5)-20}{\sqrt{18}}\right) = \Phi(2.24) = .9875.$

**c.** $P(15 \le X \le 25) = P(X \le 25) - P(X \le 14) = \Phi\left(\dfrac{(25+.5)-20}{\sqrt{18}}\right) - \Phi\left(\dfrac{(14+.5)-20}{\sqrt{18}}\right)$

$= \Phi(1.30) - \Phi(-1.30) = .9032 - .0968 = .8064.$

**60.**

**a.** $P(X \le 100) = 1 - e^{-(100)(.01386)} = 1 - e^{-1.386} = .7499.$
$P(X \le 200) = 1 - e^{-(200)(.01386)} = 1 - e^{-2.772} = .9375.$
$P(100 \le X \le 200) = P(X \le 200) - P(X \le 100) = .9375 - .7499 = .1876.$

**b.** First, since $X$ is exponential, $\mu = \dfrac{1}{\lambda} = \dfrac{1}{.01386} = 72.15$, $\sigma = 72.15$. Then
$P(X > \mu + 2\sigma) = P(X > 72.15 + 2(72.15)) = P(X > 216.45) = 1 - (1 - e^{-.01386(216.45)}) = e^{-3} = .0498.$

**c.** Remember the median is the solution to $F(x) = .5$. Use the formula for the exponential cdf and solve for $x$: $F(x) = 1 - e^{-.01386x} = .5 \Rightarrow e^{-.01386x} = .5 \Rightarrow -.01386x = \ln(.5) \Rightarrow x = -\dfrac{\ln(.5)}{.01386} = 50.01$ m.

**68.**

**a.** $E(X) = \alpha\beta = n\dfrac{1}{\lambda} = \dfrac{n}{\lambda}$; for $\lambda = .5$ and $n = 10$, $E(X) = 20$.

**b.** $P(X \le 30) = F(30/2; 10) = F(15; 10) = .930.$

**c.** $F(x; \lambda, n) = P(X \le t) = P($the $n$th event occurs in $[0, t]) = P($<u>at least</u> $n$ events occur in $[0, t])$
$= P(Y \ge n)$, where $Y =$ the number of events in $[0, t] \sim$ Poisson with parameter $\lambda t$.

Thus $F(x; \lambda, n) = 1 - P(Y < n) = 1 - P(Y \le n - 1) = 1 - \displaystyle\sum_{k=0}^{n-1} \frac{e^{-\lambda t}(\lambda t)^k}{k!}.$

**76.**

**a.** $E(X) = e^{\mu + \sigma^2/2} = e^{1.9 + .9^2/2} = e^{2.305} = 10.024.$
$V(X) = \left(e^{2(1.9)+.9^2}\right) \cdot \left(e^{.9^2} - 1\right) = 125.394 \Rightarrow \sigma_X = 11.198.$

**b.** $P(X \le 10) = \Phi\left(\dfrac{\ln(10) - 1.9}{.9}\right) = \Phi(0.447) \approx .6736.$

$P(5 \le X \le 10) = \Phi\left(\dfrac{\ln(10) - 1.9}{.9}\right) - \Phi\left(\dfrac{\ln(5) - 1.9}{.9}\right) = .6736 - \Phi(-0.32) = .6736 - .3745 = .2991.$

**84.**

**a.** $E(X) = \frac{5}{5+2} = \frac{5}{7} = .714$, $V(X) = \frac{10}{(49)(8)} = .0255$.

**b.** $f(x) = \frac{\Gamma(7)}{\Gamma(5)\Gamma(2)} \cdot x^4 \cdot (1-x) = 30\left(x^4 - x^5\right)$ for $0 \le x \le 1$, so $P(X \le .2) = \int_0^{.2} 30(x^4 - x^5)\,dx = .0016$.

**c.** Similarly, $P(.2 \le X \le .4) = \int_{.2}^{.4} 30(x^4 - x^5)\,dx = .03936$.

**d.** If $X$ is the proportion covered by the plant, then $1 - X$ is the proportion <u>not</u> covered by the plant.
$E(1 - X) = 1 - E(X) = 1 - \frac{5}{7} = \frac{2}{7} = .286$.

**86.**

**a.** $E(Y) = 10 \Rightarrow E\left(\frac{Y}{20}\right) = \frac{1}{2} = \frac{\alpha}{\alpha+\beta}$ and $V(Y) = \frac{100}{7} \Rightarrow V\left(\frac{Y}{20}\right) = \frac{100}{7(20)^2} = \frac{1}{28} = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$. After some algebra, the solutions are $\alpha = \beta = 3$.

**b.** $f(y) = 30y^2(1-y)^2$, so $P(8 \le Y \le 12) = \int_{.4}^{.6} 30y^2(1-y)^2\,dy = .365$.

**c.** We expect it to snap at 10, so $P(Y < 8 \text{ or } Y > 12) = 1 - P(8 \le Y \le 12)$
$= 1 - .365 = .665$.

**100.**

**a.** Clearly $f(x) \ge 0$. Now check that the function integrates to 1:

$$\int_0^\infty \frac{32}{(x+4)^3}\,dx = \int_0^\infty 32(x+4)^{-3}\,dx = -\frac{16}{(x+4)^2}\Bigg|_0^\infty = 0 - -\frac{16}{(0+4)^2} = 1.$$

**b.** For $x \le 0$, $F(x) = 0$. For $x > 0$,

$$F(x) = \int_{-\infty}^x f(y)dy = \int_0^x \frac{32}{(y+4)^3}dy = -\frac{1}{2}\cdot\frac{32}{(y+4)^2}\Bigg]_0^x = 1 - \frac{16}{(x+4)^2}.$$

**c.** $P(2 \le X \le 5) = F(5) - F(2) = 1 - \frac{16}{81} - \left(1 - \frac{16}{36}\right) = .247$.

**d.** $E(X) = \int_{-\infty}^\infty x \cdot f(x)dx = \int_{-\infty}^\infty x \cdot \frac{32}{(x+4)^3}dx = \int_0^\infty (x+4-4)\cdot\frac{32}{(x+4)^3}dx$

$$= \int_0^\infty \frac{32}{(x+4)^2}dx - 4\int_0^\infty \frac{32}{(x+4)^3}dx = 8 - 4 = 4 \text{ years.}$$

**e.** $E\left(\frac{100}{X+4}\right) = \int_0^\infty \frac{100}{x+4}\cdot\frac{32}{(x+4)^3}dx = 3200\int_0^\infty \frac{1}{(x+4)^4}dx = \frac{3200}{(3)(64)} = 16.67$.

## Chapter 5

**6.**

**a.** $p(4,2) = P(Y=2 \mid X=4) \cdot P(X=4) = \left[\binom{4}{2}(.6)^2(.4)^2\right] \cdot (.15) = .0518$.

**b.** $P(X=Y) = p(0,0) + p(1,1) + p(2,2) + p(3,3) + p(4,4) = .1 + (.2)(.6) + (.3)(.6)^2 + (.25)(.6)^3 + (.15)(.6)^4 =$ .4014.

**c.** $p(x,y) = 0$ unless $y = 0, 1, \ldots, x$ and $x = 0, 1, 2, 3, 4$. For any such pair,

$p(x,y) = P(Y=y \mid X=x) \cdot P(X=x) = \binom{x}{y}(.6)^y(.4)^{x-y} \cdot p_X(x)$. As for the marginal pmf of $Y$,

$p_Y(4) = P(Y=4) = P(X=4,\ Y=4) = p(4,4) = (.6)^4 \cdot (.15) = .0194$;

$p_Y(3) = p(3,3) + p(4,3) = (.6)^3(.25) + \binom{4}{3}(.6)^3(.4)(.15) = .1058$; similarly,

$p_Y(2) = p(2,2) + p(3,2) + p(4,2) = (.6)^2(.3) + \binom{3}{2}(.6)^2(.4)(.25) + \binom{4}{2}(.6)^2(.4)^2(.15) = .2678$, $p_Y(1) =$ $p(1,1) + p(2,1) + p(3,1) + p(4,1) = .3590$, and
$p_Y(0) = 1 - [.3590 + .2678 + .1058 + .0194] = .2480$.

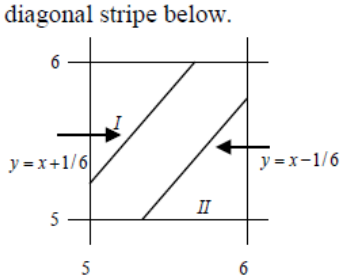
**10.**

**a.** Since $f_X(x) = \dfrac{1}{6-5} = 1$ for $5 \le x \le 6$, similarly $f_Y(y) = 1$ for $5 \le y \le 6$, and $X$ and $Y$ are independent,

$$f(x,y) = f_X(x) \cdot f_Y(y) = \begin{cases} 1 & 5 \le x \le 6, 5 \le y \le 6 \\ 0 & \text{otherwise} \end{cases}$$

**b.** $P(5.25 \le X \le 5.75,\ 5.25 \le Y \le 5.75) = P(5.25 \le X \le 5.75) \cdot P(5.25 \le Y \le 5.75)$ by independence = $(.5)(.5) = .25$.

**c.** The region $A$ is the diagonal stripe below.




$P((X,Y) \in A) = \iint_A 1\, dxdy =$ the area of $A = 1 -$ (area of $I$ + area of $II$)

$$= 1 - \left(\frac{1}{2}\cdot\frac{5}{6}\cdot\frac{5}{6} + \frac{1}{2}\cdot\frac{5}{6}\cdot\frac{5}{6}\right) = 1 - \frac{25}{36} = \frac{11}{36} = .306.$$

**12.**

**a.** $P(X > 3) = \int_3^\infty \int_0^\infty xe^{-x(1+y)}dydx = \int_3^\infty e^{-x}dx = .050.$

**b.** The marginal pdf of $X$ is $f_X(x) = \int_0^\infty xe^{-x(1+y)}dy = e^{-x}$ for $x \geq 0$. The marginal pdf of $Y$ is $f_Y(y) = \int_3^\infty xe^{-x(1+y)}dx = \dfrac{1}{(1+y)^2}$ for $y \geq 0$. It is now clear that $f(x,y)$ is not the product of the marginal pdfs, so the two rvs are not independent.

**c.** $P$(at least one exceeds 3) $= P(X > 3$ or $Y > 3) = 1 - P(X \leq 3$ and $Y \leq 3)$

$$= 1 - \int_0^3 \int_0^3 xe^{-x(1+y)}dydx = 1 - \int_0^3 \int_0^3 xe^{-x}e^{-xy}dy$$

$$= 1 - \int_0^3 e^{-x}(1 - e^{-3x})dx = e^{-3} + .25 - .25e^{-12} = .300.$$

**22.**

**a.** $E(X+Y) = \sum\sum(x+y)p(x,y) = (0+0)(.02) + (5+0)(.04) + \ldots + (10+15)(.01) = 14.10.$

Note: It can be shown that $E(X + Y)$ always equals $E(X) + E(Y)$, so in this case we could also work out the means of $X$ and $Y$ from their marginal distributions: $E(X) = 5.55$, $E(Y) = 8.55$, so $E(X + Y) = 5.55 + 8.55 = 14.10$.

**b.** For each coordinate, we need the maximum; e.g., max(0,0) = 0, while max(5,0) = 5 and max(5,10) = 10. Then calculate the sum: $E(\max(X,Y)) = \sum\sum \max(x,y)\cdot p(x,y) =$ max(0,0)(.02) + max(5,0)(.04) + … + max(10,15)(.01) = 0(.02) + 5(.04) + … + 15(.01) = 9.60.

**36.** Use the previous exercise: $\text{Cov}(X, Y) = \text{Cov}(X, aX + b) = a\text{Cov}(X, X) = aV(X) \Rightarrow$

so $\text{Corr}(X,Y) = \dfrac{aV(X)}{\sigma_X \cdot \sigma_Y} = \dfrac{aV(X)}{\sigma_X \cdot |a| \sigma_X} = \dfrac{a}{|a|} = 1$ if $a > 0$, and $-1$ if $a < 0$.

**38.**

**a.** Since each $X$ is 0 or 1 or 2, the possible values of $T_o$ are 0, 1, 2, 3, 4.
$P(T_o = 0) = P(X_1 = 0$ and $X_2 = 0) = (.2)(.2) = .04$ since $X_1$ and $X_2$ are independent.
$P(T_o = 1) = P(X_1 = 1$ and $X_2 = 0$, or $X_1 = 0$ and $X_2 = 1) = (.5)(.2) + (.2)(.5) = .20$.
Similarly, $P(T_o = 2) = .37$, $P(T_o = 3) = .30$, and $P(T_o = 4) = .09$. These values are displayed in the pmf table below.

| $t_o$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $p(t_o)$ | .04 | .20 | .37 | .30 | .09 |

**b.** $E(T_o) = 0(.04) + 1(.20) + 2(.37) + 3(.30) + 4(.09) = 2.2$. This is exactly twice the population mean: $E(T_o) = 2\mu$.

**c.** First, $E(T_o^2) = 0^2(.04) + 1^2(.20) + 2^2(.37) + 3^2(.30) + 4^2(.09) = 5.82$. Then $V(T_o) = 5.82 - (2.2)^2 = .98$. This is exactly twice the population variance: $V(T_o) = 2\sigma^2$.

**40.**

**a.** There are only three possible values of $M$: 0, 5, and 10. Let's find the probabilities associated with 0 and 10, since they're the easiest.
$P(M = 0) = P(\text{all three draws are } 0) = P(X_1 = 0) \cdot P(X_2 = 0) \cdot P(X_3 = 0) = (5/10)(5/10)(5/10) = .125$.
$P(M = 10) = P(\text{at least one draw is a } 10) = 1 - P(\underline{\text{none}} \text{ of the three draws is a } 10) =$
$1 - P(X_1 \neq 10) \cdot P(X_2 \neq 10) \cdot P(X_3 \neq 10) = 1 - (8/10)(8/10)(8/10) = .488$.
Calculating all the options for $M = 5$ would be complicated; however, the three probabilities must sum to 1, so $P(M = 5) = 1 - [.125 + .488] = .387$. The probability distribution of $M$ is displayed in the pmf table below.

| $m$ | 0 | 5 | 10 |
|---|---|---|---|
| $p(m)$ | .125 | .387 | .488 |

An alternative solution would be to list all 27 possible combinations using a tree diagram and computing probabilities directly from the tree.

**48.**

**a.** No, it doesn't seem plausible that waist size distribution is approximately normal. The normal distribution is symmetric; however, for this data the mean is 86.3 cm and the median is 81.3 cm (these should be nearly equal). Likewise, for a symmetric distribution the lower and upper quartiles should be equidistant from the mean (or median); that isn't the case here.

If anything, since the upper percentiles stretch much farther than the lower percentiles do from the median, we might suspect a right-skewed distribution, such as the exponential distribution (or gamma or Weibull or …) is appropriate.

**b.** Irrespective of the population distribution's shape, the Central Limit Theorem tells us that $\bar{X}$ is (approximately) normal, with a mean equal to $\mu = 85$ cm and a standard deviation equal to $\sigma / \sqrt{n} = 15 / \sqrt{277} = .9$ cm. Thus,

$$P(\bar{X} \geq 86.3) = P\left(Z \geq \frac{86.3 - 85}{.9}\right) = 1 - \Phi(1.44) = .0749$$

**c.** Replace 85 with 82 in (b):

$$P(\bar{X} \geq 86.3) = P\left(Z \geq \frac{86.3 - 82}{.9}\right) = 1 - \Phi(4.77) \approx 1 - 1 = 0$$

That is, if the population mean waist size is 82 cm, there would be almost no chance of observing a sample mean waist size of 86.3 cm (or higher) in a random sample if 277 men. Since a sample mean of 86.3 was actually observed, it seems incredibly implausible that $\mu$ would equal 82 cm.

**52.** We have $X \sim N(10,1)$, $n = 4$, $\mu_{T_o} = n\mu = (4)(10) = 40$ and $\sigma_{T_o} = \sigma\sqrt{n} = 2$. Hence, $T_o \sim N(40, 2)$. We desire the 95th percentile of $T_o$: $40 + (1.645)(2) = 43.29$ hours.

**56.**

**a.** Let $X$ = the number of erroneous bits out of 1000, so $X \sim \text{Bin}(1000, .10)$. If we approximate $X$ by a normal rv with $\mu = np = 100$ and $\sigma^2 = npq = 90$, then with a continuity correction $P(X \le 125) = P(X \le 125.5) \approx P\left(Z \le \frac{125.5-100}{\sqrt{90}}\right) = P(Z \le 2.69) = \Phi(2.69) = .9964$.

**b.** Let $Y$ = the number of errors in the second transmission, so $Y \sim \text{Bin}(1000, .10)$ and is independent of $X$. To find $P(|X - Y| \le 50)$, use the facts that $E[X - Y] = 100 - 100 = 0$ and $V(X - Y) = V(X) + V(Y) = 90 + 90 = 180$. So, using a normal approximation to both binomial rvs, $P(|X - Y| \le 50) \approx P\left(|Z| \le \frac{50}{\sqrt{180}}\right) = P(|Z| \le 3.73) \approx 1$.

**58.**

**a.** $E(27X_1 + 125X_2 + 512X_3) = 27E(X_1) + 125E(X_2) + 512E(X_3)$
$= 27(200) + 125(250) + 512(100) = 87{,}850$.
$V(27X_1 + 125X_2 + 512X_3) = 27^2\, V(X_1) + 125^2\, V(X_2) + 512^2\, V(X_3)$
$= 27^2\,(10)^2 + 125^2\,(12)^2 + 512^2\,(8)^2 = 19{,}100{,}116$.

**b.** The expected value is still correct, but the variance is not because the covariances now also contribute to the variance.

**62.** $E(X_1 + X_2 + X_3) = E(X_1) + E(X_2) + E(X_3) = 15 + 30 + 20 = 65$ min, and
$V(X_1 + X_2 + X_3) = 1^2 + 2^2 + 1.5^2 = 7.25 \Rightarrow SD(X_1 + X_2 + X_3) = 2.6926$ min.
Thus, $P(X_1 + X_2 + X_3 \le 60) = P\left(Z \le \frac{60-65}{2.6926}\right) = P(Z \le -1.86) = .0314$.

**68.** Let $X_1$ and $X_2$ denote the (constant) speeds of the two planes.

**a.** After two hours, the planes have traveled $2X_1$ km and $2X_2$ km, respectively, so the second will not have caught the first if $2X_1 + 10 > 2X_2$, i.e. if $X_2 - X_1 < 5$.
$X_2 - X_1$ has a mean $500 - 520 = -20$, variance $100 + 100 = 200$, and standard deviation 14.14. Thus,
$P(X_2 - X_1 < 5) = P\left(Z < \frac{5-(-20)}{14.14}\right) = P(Z < 1.77) = .9616$.

**b.** After two hours, #1 will be $10 + 2X_1$ km from where #2 started, whereas #2 will be $2X_2$ from where it started. Thus, the separation distance will be at most 10 if
$|2X_2 - 10 - 2X_1| \le 10$, i.e. $-10 \le 2X_2 - 10 - 2X_1 \le 10$ or $0 \le X_2 - X_1 \le 10$. The corresponding probability is $P(0 \le X_2 - X_1 \le 10) = P(1.41 \le Z \le 2.12) = .9830 - .9207 = .0623$.

**74.** $X$ is approximately normal with $\mu_1 = (50)(.7) = 35$ and $\sigma_1^2 = (50)(.7)(.3) = 10.5$, as is $Y$ with $\mu_2 = 30$ and $\sigma_2^2 = 12$. Thus $\mu_{X-Y} = 5$ and $\sigma_{X-Y}^2 = \sigma_X^2 + \sigma_Y^2 = 22.5$, so
$P(-5 \le X - Y \le 5) \approx P\left(\frac{-10}{4.74} \le Z \le \frac{0}{4.74}\right) = P(-2.11 \le Z \le 0) = .4826$.

## Chapter 6

**2.**

**a.** A sensible point estimate of the population mean $\mu$ is the sample mean, $\bar{x} = 49.95$ mg/dl.

**b.** Of interest is the population median, $\tilde{\mu}$. The logical point estimate is the sample median, $\tilde{x}$, which is the average of the 10th and 11th ordered values: $\tilde{x} = 47.5$ mg/dl.

**c.** The point estimate of the population sd, $\sigma$, is the sample standard deviation, $s = 16.81$ mg/dl.

**d.** The natural estimate of $p = P(X \geq 60)$ is the sample proportion of HDL observations that are at least 60. In this sample of $n = 20$ observations, 4 are 60 or higher, so the point estimate is $\hat{p} = 4/20 = .2$.

**8.**

**a.** With $p$ denoting the true proportion of *non*-defective components, $\hat{p} = \dfrac{80-12}{80} = .85$.

**b.** $P(\text{system works}) = p^2$, so an estimate of this probability is $\hat{p}^2 = \left(\dfrac{68}{80}\right)^2 = .723$.

**10.**

**a.** The hint tells us that $E(\bar{X}^2) = V(\bar{X}) + [E(\bar{X})]^2$. We know that $E(\bar{X}) = \mu$ and $SD(\bar{X}) = \dfrac{\sigma}{\sqrt{n}}$, so

$E(\bar{X}^2) = \left(\dfrac{\sigma}{\sqrt{n}}\right)^2 + [\mu]^2 = \dfrac{\sigma^2}{n} + \mu^2$. Since $E(\bar{X}^2) \neq \mu^2$, we've discovered that $\bar{X}^2$ is <u>not</u> an unbiased estimator of $\mu^2$!

In fact, the bias equals $E(\bar{X}^2) - \mu^2 = \dfrac{\sigma^2}{n} > 0$, so $\bar{X}^2$ is *biased high*. It will tend to *over*estimate the true value of $\mu^2$.

**b.** By linearity of expectation, $E(\bar{X}^2 - kS^2) = E(\bar{X}^2) - kE(S^2)$. The author proves in Section 6.1 that $E(S^2) = \sigma^2$, so $E(\bar{X}^2) - kE(S^2) = \dfrac{\sigma^2}{n} + \mu^2 - k\sigma^2$.

The goal is to find $k$ so that $E(\bar{X}^2 - kS^2) = \mu^2$. That requires $\dfrac{\sigma^2}{n} - k\sigma^2 = 0$, or $k = \dfrac{1}{n}$.

**22.**

**a.** $E(X)=\int_0^1 x(\theta+1)x^{\theta}dx=\frac{\theta+1}{\theta+2}=1-\frac{1}{\theta+2}$, so the moment estimator $\hat{\theta}$ is the solution to $\overline{X}=1-\frac{1}{\hat{\theta}+2}$, yielding $\hat{\theta}=\frac{1}{1-\overline{X}}-2$. Since $\overline{x}=.80, \hat{\theta}=5-2=3$.

**b.** $f(x_1,\ldots,x_n;\theta)=(\theta+1)^n(x_1x_2\ldots x_n)^{\theta}$, so the log likelihood is $n\ln(\theta+1)+\theta\sum\ln(x_i)$. Taking $\frac{d}{d\theta}$ and equating to 0 yields $\frac{n}{\theta+1}=-\sum\ln(x_i)$, so $\hat{\theta}=-\frac{n}{\sum\ln(X_i)}-1$. Taking $\ln(x_i)$ for each given $x_i$ yields ultimately $\hat{\theta}=3.12$.

**24.** The number of incorrect diagnoses, $X$, until $r$ correct diagnoses are made has a negative binomial distribution: $X \sim \text{nb}(r, p)$. To find the mle of $p$, we'll take the derivative of the log-likelihood function $\ell(p)=\ln\left[\binom{x+r-1}{x}p^r(1-p)^x\right]=\ln\binom{x+r-1}{x}+r\ln(p)+x\ln(1-p)$, set it equal to zero, and solve for $p$.

$\ell'(p)=\frac{d}{dp}\left[\ln\binom{x+r-1}{x}+r\ln(p)+x\ln(1-p)\right]=\frac{r}{p}-\frac{x}{1-p}=0 \Rightarrow r(1-p)=xp \Rightarrow p=r/(r+x)$, so the mle of $p$ is $\hat{p}=\frac{r}{r+x}$. This is the number of successes over the total number of trials; with $r = 3$ and $x = 17$, $\hat{p}=.15$.

Yes, this is the same as the mle based on a random sample of 20 mechanics with 3 correct diagnoses — see the binomial mle for $p$ in Exercise 6.20.

Both mles are equal to the fraction (number of successes) / (number of trials).

In contrast, the unbiased estimator from Exercise 6.17 is $\hat{p}=\frac{r-1}{r+x-1}$, which is <u>not</u> the same as the maximum likelihood estimator. (With $r = 3$ and $x = 17$, the calculated value of the unbiased estimator is 2/19, rather than 3/20.)